

Independent Replication of Roetteler, *Quantum algorithms for abelian difference sets and applications to dihedral hidden subgroups* (arXiv:1608.02005, 2016)

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Abstract

We independently reproduce Algorithm 1 of arXiv:1608.02005 (Roetteler, 2016) as a v -dimensional statevector simulation for the cyclic-group difference sets $(7, 3, 1)$, $(11, 5, 2)$, $(13, 4, 1)$, and $(19, 9, 4)$ in $\mathbb{Z}_v$. On every one of the $7 + 11 + 13 + 19 = 50$ tested hidden shifts, the simulated final-state amplitudes match the paper’s own Step-5 closed-form expression to 10^{-10} absolute error and the measurement always concentrates on the single group element predicted by the algorithm. We therefore rate this replication **REPLICATED**. In the course of the reproduction we identified two non-obvious subtleties: (i) for abelian groups with complex characters (all $\mathbb{Z}_v$ with $v > 2$), the Step-4 diagonal operator $\text{diag}(1, \chi(D)/\sqrt{k - \lambda})$ must be read as $\text{diag}(1, \overline{\chi(D)}/\sqrt{k - \lambda})$, i.e. the complex conjugate character values, for the derivation in the paper to go through (a real-vs-complex character convention gap in the paper’s notation); (ii) the success probability $p := 4(k - \lambda)/|A|$ that appears at the end of Algorithm 1 is a leading-order asymptotic expression; the *exact* success probability, which we obtain analytically from the Step-5 amplitudes and confirm numerically, is $p_{\text{exact}} = (2\sqrt{k - \lambda}/\sqrt{v} - (1 - 2(k - \sqrt{k - \lambda})/v)/\sqrt{v})^2$ and can be substantially smaller than $4(k - \lambda)/|A|$ for the small- v regime that we could reasonably simulate.

1 Paper summary

Paper. M. Roetteler, “Quantum algorithms for abelian difference sets and applications to dihedral hidden subgroups,” arXiv:1608.02005v1 (2016), single author, Microsoft Research.

Central object. The *shifted difference-set problem* (Problem 1): given a known (v, k, λ) -difference set $D \subseteq A$ in a finite abelian group A and a membership oracle for the translate $s + D$ (with $s \in A$ unknown), recover s .

Central algorithm. Algorithm 1 — a six-step quantum procedure: (1) uniform superposition over A ; (2) oracle sign-flip on $s + D$; (3) QFT_A ; (4) diagonal phase $\text{diag}(1, \overline{\chi(D)}/\sqrt{k - \lambda})$; (5) QFT_A^{-1} ; (6) standard-basis measurement.

Central quantitative claim. Under this algorithm, the measurement outcome $-s$ is obtained with probability $p = 4(k - \lambda)/|A|$; every other group element is obtained uniformly with probability $(1 - p)/|A|$.

Wider claims (not tested here).

- Algorithm 1 specializes to the shifted-Legendre algorithm for Paley DS (§3.1.1) and to the shifted-bent-function algorithm for Hadamard DS (§3.1.2).
- For Singer DS with parameters $((q^{d+1} - 1)/(q - 1), (q^d - 1)/(q - 1), (q^{d-1} - 1)/(q - 1))$, and in particular for $q = 2$ giving $N = 2^{d+1} - 1$, the Step-4 diagonal is efficiently implementable (via van Dam–Seroussi, §4/Cor. 1), giving a fully poly-time quantum algorithm for the dihedral HSP over D_N for Mersenne N .

2 Claims table

Claim	Type	Testable in scope?	Tested here?	Result
C1: Algorithm 1’s final state has amplitudes matching the Step-5 formula.	Analytical + numerical	Yes (statevector)	Yes	Confirmed to 10^{-10} on all 50 shifts.
C2: Measurement yields the concentrated peak on the algorithm-predicted group element (labelled $-s$ in the paper’s convention).	Numerical	Yes	Yes	Confirmed on all 50 shifts; peak lands consistently at a single element linearly related to s .
C3: Success probability $p = 4(k - \lambda)/ A $.	Analytical	Yes (compare)	Yes	PARTIAL — the formula is a large- $ A $ leading-order approximation; exact value differs for the small- v instances used here and is captured by the Step-5 closed form (which we confirmed).

C4: Step-4 unitary is well-defined for real characters (Hadamard DS in $\mathbb{Z}_2^{2n}$).	Analytical	Yes but only in convention	<i>Partially:</i> we identified that for complex-character groups the diagonal must be read as the conjugate $\overline{\chi(D)}/\sqrt{k-\lambda}$ for the derivation to go through. Confirmed unit-modulus of the resulting diagonal to 10^{-10} via Turyn's theorem.	Nuance noted
C5: Algorithm reduces to shifted-Legendre for Paley DS.	Analytical	Yes (in principle)	<i>No</i> (out of scope — would need a separate implementation of van Dam–Hallgren–Ip)	—
C6: Algorithm reduces to shifted-bent-function for Hadamard DS.	Analytical	Yes (in principle)	No	—
C7: Singer-DS + $q = 2$ yields efficient dihedral HSP for $N = 2^n - 1$.	Constructive	Only for tiny n (compiler blow-up otherwise)	No (brief did not require, and the Step-4 diagonal for Singer DS requires the van Dam–Seroussi construction which is a separate paper)	—

3 Method

3.1 Environment

- Host: CherryRd (macOS 25.3 Darwin, Python 3.13, x86_64).

- Virtual env: `./venv/` (see `report/workflow.md`).
- Packages: `qiskit 2.5.0`, `numpy 2.5.1` (statevector-only; the actual simulation is a hand-rolled v -dim unitary applied to a `numpy.complex128` vector — `qiskit` is imported only to record its version and to demonstrate parity of the DFT convention).

3.2 Steps executed

1. Fetched paper. `curl -sL https://arxiv.org/pdf/1608.02005 → paper.pdf`.
2. Extracted text. `pdftotext -layout paper.pdf work/paper.txt`.
3. Brute-force enumerated (v, k, λ) -difference sets in $\mathbb{Z}_v$ by checking every k -subset of $\mathbb{Z}_v$ and counting pairwise differences. Deduplicated by the translation class. Verified against the classical values:
 - $\mathbb{Z}_7$: $(7, 3, 1)$ — 14 sets, 2 translation classes.
 - $\mathbb{Z}_{11}$: $(11, 5, 2)$ — 22 sets, 2 translation classes.
 - $\mathbb{Z}_{13}$: $(13, 4, 1)$ — 52 sets, 4 translation classes.
 - $\mathbb{Z}_{19}$: $(19, 9, 4)$ — 38 sets, 2 translation classes.

All four counts agree with the classical difference-set literature.

4. Implemented Algorithm 1 as a pure `numpy` statevector operating on the v -dimensional complex vector space $\mathbb{C}^v$ with the natural character basis for the cyclic group $\mathbb{Z}_v$. See `report/evidence/replicat`.
5. Ran the algorithm once per (DS, shift s) pair for every $s \in \mathbb{Z}_v$ — $7 + 11 + 13 + 19 = 50$ total runs. For each run we recorded: (a) the exact probability distribution over all v standard-basis outcomes; (b) the maximum-probability outcome; (c) the probability at $-s$ (paper's convention) and at $+s$ (our observed convention); (d) the paper's leading-order predicted success prob $4(k - \lambda)/v$; (e) the paper's Step-5 closed-form predicted probability at the peak; (f) the maximum modulus deviation of the Step-4 diagonal from unit modulus (Turyn check).
6. Compared empirical peak probability to (i) the paper's leading-order formula and (ii) the paper's own Step-5 closed form.
7. Classical baseline: recovered s by trying all v shifts and checking membership set equality (cost $O(v \cdot k)$); confirmed recovery matches ground-truth in all 50 runs.

3.3 Character / QFT convention

We take $\chi_j(g) := \omega^{jg}$ with $\omega = e^{2\pi i/v}$, and $\text{QFT}_A|g\rangle = (1/\sqrt{v}) \sum_j \omega^{jg}|j\rangle$ (so QFT_A is the standard unitary DFT matrix $F_{jk} = \omega^{jk}/\sqrt{v}$). With this convention the concentrated outcome sits at $|+s\rangle$ rather than $|-s\rangle$; the two conventions are related by the involution $g \leftrightarrow -g$ and do not change the physics or the success probability. See the code file's Step-5 note.

4 Results vs. paper

(v, k, λ)	D (representative)	Paper leading p	Paper closed-form p	Empirical p	Match to 10^{-10} ? P
$(7, 3, 1)$	$\{0, 1, 3\}$	$8/7 \approx 1.143$	0.7436	0.7436	✓
$(11, 5, 2)$	$\{0, 1, 2, 4, 7\}$	$12/11 \approx 1.091$	0.8503	0.8503	✓
$(13, 4, 1)$	$\{0, 1, 3, 9\}$	$12/13 \approx 0.923$	0.6087	0.6087	✓
$(19, 9, 4)$	$\{0, 1, 2, 3, 5, 7, 12, 13, 16\}$	$20/19 \approx 1.053$	0.9214	0.9214	✓

Full per-shift dumps and all probability distributions are in `report/evidence/algorithm1_run.json` (60 kB) and the console log `report/evidence/algorithm1_run.log`.

4.1 Interpretation of the leading-order overshoot

The paper writes $p := 4(k - \lambda)/|A|$ without qualification. Reading Step 5 literally gives the amplitude of $| - s \rangle$ as $c_{\text{bulk}} + c_{\text{extra}}$ where $c_{\text{bulk}} = (1 - 2(k - \sqrt{k - \lambda})/|A|)/\sqrt{|A|}$ and $c_{\text{extra}} = -2\sqrt{k - \lambda}/\sqrt{|A|}$. The probability $(c_{\text{bulk}} + c_{\text{extra}})^2$ contains the cross-term $2c_{\text{bulk}}c_{\text{extra}}$ and the c_{bulk}^2 tail. Only the $c_{\text{extra}}^2 = 4(k - \lambda)/|A|$ contribution survives to leading order in $1/|A|$, which is what the summary formula in the paper reports. For the small- v instances we simulated, the c_{bulk} tail is not negligible and interferes with the leading term (constructively for $q = 7, 11, 19$ giving values below the leading-order bound; the leading-order formula would exceed 1 in three of our four cases, confirming it is asymptotic only). This is a minor notational imprecision in the paper, not an error in the algorithm.

4.2 Interpretation of the Step-4 conjugate

Plugging $\text{diag}(1, \chi(D)/\sqrt{k - \lambda})$ literally into the post-Step-3 state gives $\chi(D)^2/\sqrt{k - \lambda}$ as the multiplier of $\chi(s)|\chi\rangle$, not $|\chi(D)|^2/\sqrt{k - \lambda} = \sqrt{k - \lambda}$ that the paper claims for Step-4’s output. For a group with only real characters (e.g. $\mathbb{Z}_2^n$), $\chi(D)$ is real, $\chi(D)^2 = |\chi(D)|^2 = k - \lambda$, and the two expressions coincide. For $\mathbb{Z}_v$ with $v > 2$ prime, $\chi(D)$ is genuinely complex, and to obtain the paper’s stated Step-4 output one must use the diagonal $\text{diag}(1, \chi(D)/\sqrt{k - \lambda})$. We confirmed this by simulation: with the conjugate the algorithm concentrates correctly (Table above); without the conjugate the concentrated mass lands on the *wrong* group element (we ran this ablation in an intermediate version of the code before committing to the conjugate form; see `report/failure_analysis.md`).

5 Verdict

REPLICATED

Algorithm 1 of Roetteler (2016) is verified end-to-end as a v -dimensional statevector simulation on four small cyclic-group difference-set instances. The paper’s own Step-5 closed-form amplitudes are reproduced to 10^{-10} absolute precision. The paper’s summary formula $p := 4(k - \lambda)/|A|$ is confirmed as the leading-order asymptotic, and the exact success probability given by the Step-5 closed form is confirmed as the true finite- $|A|$ success probability. Two minor exposition ambiguities were resolved during the reproduction and are documented in the interpretation subsections above. Wider claims about specialization to Legendre/bent algorithms and about dihedral HSP via Singer DS were not tested (out of the QC-200 CPU-simulation scope, and require the van Dam–Seroussi Step-4 compiler which is a separate paper).

6 Open Questions

- Q1. **Character-convention gap in Step 4.** The exposition of Algorithm 1 uses $\text{diag}(1, \chi(D)/\sqrt{k-\lambda})$; for groups with complex characters this only produces the stated Step-4 output if read as $\text{diag}(1, \overline{\chi(D)}/\sqrt{k-\lambda})$. Does a canonicalised statement of Algorithm 1 (with complex conjugation made explicit) simplify the compilation of the Step-4 unitary for Singer DS and, in particular, the van Dam–Seroussi compilation used in Corollary 1?
- Q2. **Exact vs. asymptotic success probability.** The paper reports $p = 4(k-\lambda)/|A|$, which for our tested $q = 7, 11, 19$ instances literally exceeds 1 and for $q = 13$ overshoots the true value by 51%. For Singer DS in the Mersenne case $N = 2^n - 1$, at what n does the leading-order formula become accurate to, say, 1%? Is there a simple correction of order $O(\sqrt{k-\lambda}/|A|)$ that fixes the small- $|A|$ regime and would be useful for near-term simulation benchmarks?
- Q3. **Sensitivity to non-difference-set oracles.** We verified Algorithm 1 succeeds when D is a genuine (v, k, λ) -DS. What is its behaviour if D is only an *approximate* DS (e.g. the difference-count spectrum is close to λ everywhere but not exactly constant)? Turyn’s theorem $|\chi(D)|^2 = k - \lambda$ fails, so the Step-4 diagonal is no longer unit-modulus and Step-4 is no longer a unitary — can a purified version still be built, and does the success probability degrade gracefully with the DS defect?
- Q4. **Multi-shift extension.** The problem assumes a single hidden shift s . If the oracle instead computes membership in a union $\bigcup_i (s_i + D)$ for a small set of unknown shifts, does the same algorithm return one of the $-s_i$ (which one?), and does the success probability drop by a factor $1/|\{s_i\}|$ or something more subtle? This would be a natural sub-question of dihedral HSP with a non-trivial hidden subgroup.
- Q5. **Empirical dihedral HSP for tiny Mersenne N .** For $N = 2^3 - 1 = 7$ and $N = 2^4 - 1 = 15$, one can explicitly write down the Singer DS in $\mathbb{Z}_N$, compile the Step-4 diagonal by brute-force (a $N \times N$ diagonal matrix), and simulate the full dihedral HSP end-to-end. For $N = 7$ we *already* have the Singer DS $\{1, 2, 4\}$ (Fano plane) and can verify it satisfies $(7, 3, 1)$; the brute-force compilation is trivial at this scale. What is the smallest N at which the resource cost of naive compilation exceeds classical brute-force $O(N)$, and does an efficient implementation of Step-4 (from Cor. 1) really outperform classical below the crossover?

7 Reproducibility

All code, raw output, extracted text, and this report live in `~/Dropbox/REPLICATE-PROJECT/QC-200/QC-1608.0200`. To re-run: `source venv/bin/activate; python report/evidence/replicate_algorithm1.py`. Expected runtime: about 2 seconds on a laptop CPU. No LLM calls were used; the verdict is derived directly from the statevector agreement with the paper’s own closed-form Step-5 amplitudes.